SAR Automatic Target Recognition Method Based on Multi-Stream Complex-Valued Networks

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***Abstract*— In synthetic aperture radar automatic target recognition (SAR-ATR), target information is usually propagated and reserved in complex-valued form, namely, magnitude information and phase information. However, most of the existing SAR target recognition methods only focus on real-valued (magnitude information) calculations and ignore the phase information of targets, yielding poor recognition performance. To overcome this limitation, this article proposes a multi-stream (MS) feature fusion SAR target recognition method based on complex-valued operations, called MS complex-valued networks (MS-CVNets), to utilize the phase information of the target effectively. First of all, a series of complex-valued operation blocks are constructed to satisfy the network training in the complex field, such as complex convolution, complex batch normalization, complex activation, complex pooling, and complex full connection. Besides, an MS structure is employed by applying different convolution kernels to extract multiscale information of targets, further enhancing the representation ability of the model. Experimental results on the moving and stationary target acquisition and recognition (MSTAR) dataset illustrate that compared with the current state-of-the-art real-valued based models, MS-CVNets can achieve better recognition results under both standard operating conditions (SOCs) and extended operating conditions (EOCs), validating the effectiveness and superiority of the proposed method.**

***Index Terms*— Automatic target recognition (ATR), complex-valued network (CVNet), deep learning (DL), feature fusion, multi-stream (MS) structure, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has received an increasing attention in many application scenarios, such as military, geology, and natural disaster monitoring [1]–[5], due to its all-day and all-weather remote sensing data acquisition capabilities. In recent years, deep learning (DL) networks represented by convolutional neural network (CNN), generative adversarial network (GAN), and so on have achieved excellent performance for a variety of tasks, including in computer vision and natural language processing and text translation, greatly promoting the development of intelligent information extraction [6]–[10]. Inspired by the success of DL technology, researchers attempt to introduce DL into SAR image interpretation tasks to enhance data representation ability. For example, Xie *et al.* [11] pioneered in applying DL to the task of polarimetric SAR (PolSAR) terrain surface image classification, by integrating a stacked sparse autoencoder (SAE) to automatically extract the hierarchical features of the PolSAR image. Compared with the traditional feature extraction approaches, SAE can possess better capability of learning the high-dimensional representation, which greatly improved the classification accuracy. Geng *et al.* [12] proposed a deep supervision and contraction neural network (DSCNN) for SAR image classification, and a histogram of oriented gradient (HOG) descriptor is designed to effectively obtain the information between terrain surface features and labels in a supervision penalty mechanism. Moreover, due to the lack of real training data, Song *et al.* [13] studied a DCNN rough surface parameter inversion method based on simulation data, which is able to invert the required parameters with reasonable accuracy.

For the target detection and recognition task, Chen *et al.* [14] executed a deep neural network, called A-ConvNet, to the SAR automatic target recognition (SAR-ATR) task for the first time, achieving a breakthrough target recognition accuracy (99.13%) on the ten categories (SOC-10) dataset of moving and stationary target acquisition and recognition (MSTAR). Cui *et al.* [15] applied the spatial shuffle-group enhance attention to the detection and recognition of ship targets in large scenes, which effectively reduced the false alarm rate of targets. To solve the bottleneck of target recognition under the condition of limited samples, Fu *et al.* [16] proposed a meta-learning framework to achieve target few-shot learning (FSL). The proposed learning framework is mainly composed of a metalearner and a base learner, which can effectively learn the target features and the appropriate update strategy. Concerning unknown SAR targets identification, Zeng *et al.* [17] proposed a novel identification method based on the feature extraction network and Kullback-Leibler divergence and relative position angle (KLD-RPA) joint discrimination, called Fea-DA, which can not only ensure high-precision known target recognition but also have excellent performance in discriminating unknown targets. Furthermore, for the purpose of understanding the decision-making mechanism of DL

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in the SAR target recognition process, Belloni *et al.* [18] analyzed the impact of targets, shadows, and backgrounds on the recognition performance of deep CNNs. The experimental results demonstrate that the type and orientation of the SAR target will dramatically affect the activation state of the target feature in CNN. Wang *et al.* [19] introduced visual attention mechanism to solve the problems of low accuracy and slow speed for traditional target detection methods in large scenes. Although these DL-based methods have achieved superior performance in the research of SAR image interpretation, they only consider that real-value-based calculations, such as the magnitude information in the SAR image, lack sufficient exploration of phase information.

In general, the SAR image data output by the imaging system are essentially preserved in complex-valued form [20]–[22], which represents the complex scattering coefficient of the scattering centers of the target, and its phase value reflects the response of these scattering centers to the phase of the incident electromagnetic wave. Meanwhile, since the artificial target contains a large number of scattering centers composed by small standard scatterers, its complex scattering coefficient is closely related to the structural characteristics of the target. Therefore, the phase contains a wealth of information about the target, such as shape, edge, orientation, and reflectivity. From the perspective of signal processing, the utilization of phase information can effectively improve the information completeness of SAR targets, making it possible for subsequent target processing with higher task requirements. For instance, according to the study of [23], the phase information is capable of promoting the intelligibility of the speech signal. Oppenheim and Lim [24] investigated the amplitude coding recovering method by introducing the phase information, and the experimental results verified its effectiveness. For the SAR-ATR task, considering both the magnitude information and phase information of the target is of great significance to the recognition performance of SAR targets. To effectively utilize the phase information of SAR targets, Arjovsky *et al*. [25] and Wisdom *et al.* [26] provided fundamental theory and mathematical basis for the application of complex numbers in deep CNNs. Trabelsi *et al.* [27] proposed a general architecture for the construction of DL networks based on complex-valued computation and successfully applied to computer vision and speech signal processing tasks. Zhang *et al.* [28] explored the feasibility of the complex-valued CNN (CV-CNN) in PolSAR images and achieved satisfying results of ground object classification on the Flevoland and Oberpfaffenhfen datasets. Zhang *et al.* [29] presented a CNN model based on fast complex-valued calculations to identify jamming signals in radar systems. The experimental results revealed that the designed complex-valued calculation model could achieve better radar jamming signal recognition performance than the traditional real-valued calculation model. Moore *et al.* [30] studied the influence of the phase information in the SAR target on the estimation of the scattering parameters, demonstrating that the parameter estimation after considering the phase information can provide a more compact Cramer–Rao lower bound (CRLB). Argenti *et al.* [31] analyze the effects of signal and noise model, signal representation, estimation theory, and evaluation quality on SAR image despeckling. Recent research validates the feasibility of applying phase information in SAR-ATR [32], [33], but there is still numerous work to be carried out for improvement in effectively integrating the DL-based SAR-ATR methods with the phase information, such as the production and labeling of complex-valued SAR datasets, tools, and general architectures for deep CNNs based on complex-valued calculations.

To solve the training and learning problems of complex-valued parameters in deep CNNs and enhance the recognition performance of the target recognition model in the SAR-ATR task, this article proposes a multi-stream (MS) feature fusion CNNs based on complex-valued calculations, called MS complex-valued networks (MS-CVNets), by fully considering the magnitude information and phase information in the complex-valued SAR target to improve the representation ability of the DL model for target features. In addition, a complex-valued SAR target dataset based on the MSTAR dataset [34]–[37] with both real and imaginary description is used for the analysis and evaluation of the model. More specifically, the main contributions of this article can be summarized as follows.

1) This article proposes two pooling methods based on complex number operations, complex-valued maximum pooling (CMax-Pooling) and complex-valued average pooling (CAvg-Pooling), to solve the mismatch problem in the complex field and reduce information loss in complex-valued SAR targets.
2) The MS complex-valued convolution (CConv) architecture is employed by introducing different sizes of convolution kernels to extract the multiscale information in the scene, further enhancing the representation ability of the target.
3) A series of feature fusion algorithms based on complex-valued calculations for SAR targets, such as complex-valued average fusion (CAF), complex-valued maximum fusion (CMF), and complex-valued concatenation fusion (CCF), are designed to effectively fusion MS target features.

The remaining content of this article is arranged as follows. Section II is the complex-value operation building blocks, including the architecture and principle of CConv, complex-valued batch normalizaton (CBN), and other modules. Section III details the proposed MS-CVNets. The experimental results of MS-CVNets are analyzed and discussed in Section IV. Section V is the summary and outlook of this article.

II. Complex-Valued Operation Building Blocks

In view of the lack of a general tool and framework based on complex-valued calculation in SAR-ATR, following by the work of [27], this section mainly introduces the complex-valued operation building blocks in MS-CVNets in detail.

In general, a complex number can be expressed as $z = U \cdot e^{i\theta} = a + ib$, where U represents the magnitude, θ denotes the phase angle, in the polar coordinates, and a and b are the real and imaginary parts in rectangular coordinates, respectively. Therefore, we can easily derive the complex-valued

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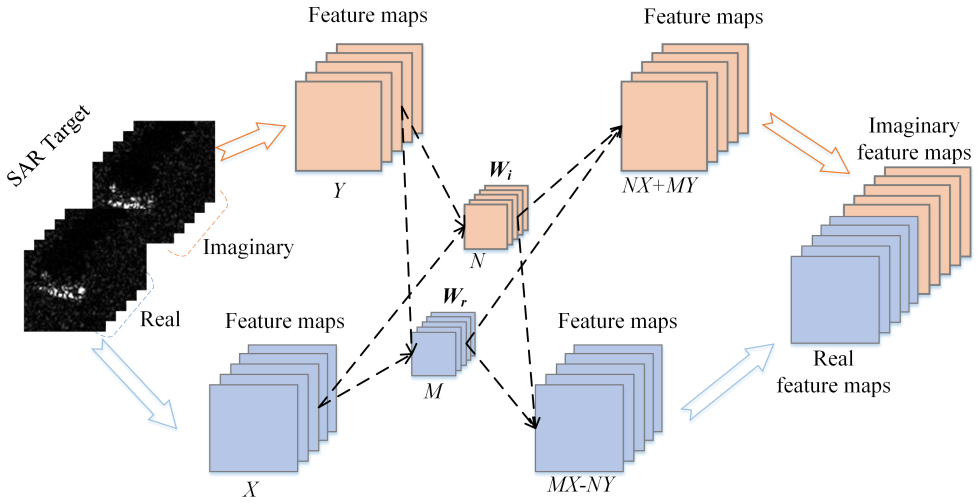

Fig. 1. Schematic of the proposed CConv.

representation of the SAR image in the two-dimensional space $\mathbf{Z} = \mathbf{U} \cdot e^{i\boldsymbol{\theta}} = \mathbf{A} + i \cdot \mathbf{B}$, where $\mathbf{U}$ and $\boldsymbol{\theta}$ indicate the magnitude and phase information of the SAR target, respectively, and $\mathbf{A}$ and $\mathbf{B}$ are the real and imaginary parts of the target, respectively.

A. Complex-Valued 2-D-Convolution

In order to realize the convolution operation of complex numbers, we carry out the simulation of the complex-valued operations in the real-valued space with the magnitude information and phase information of the SAR target. The schematic of the CConv operation of SAR image is shown in Fig. 1. First, define the complex kernel matrix of the $l+1$th convolutional layer in the Cartesian coordinate system $\mathbf{W} = \mathbf{M} + i \cdot \mathbf{N}$, and the output of the lth convolutional layer $\mathbf{H}^l = \mathbf{X} + i \cdot \mathbf{Y}$, and the convolution calculation result of the $l+1$th layer can be written as follows:

$$\begin{aligned} \mathrm{CConv}\left(\mathbf{O}^{l+1}\right) &= \mathbf{W} * \mathbf{H} + b^{l+1} \\ &= (\mathbf{M} * \mathbf{X} - \mathbf{N} * \mathbf{Y}) + i \cdot (\mathbf{N} * \mathbf{X} + \mathbf{M} * \mathbf{Y}) + b^{l+1} \end{aligned} \tag{1}$$

where b^{l+1} is the computation bias, and $b^{l+1} = b_r + i \cdot b_i$. By transforming the real and imaginary components of CConv into the matrix operation form, we have

$$\begin{bmatrix} \Re\left(\mathbf{O}^{l+1}\right) \\ \Im\left(\mathbf{O}^{l+1}\right) \end{bmatrix} = \begin{bmatrix} \mathbf{M} & -\mathbf{N} & b_r \\ \mathbf{N} & \mathbf{M} & b_i \end{bmatrix} * \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \\ 1 \end{bmatrix}. \tag{2}$$

B. Complex-Valued Batch Normalization

Batch normalization (BN) is a common and effective model optimization technique in DL, which can efficiently accelerate the convergence of the network model and avoid training problems to some extent, such as gradient disappearance and gradient explosion. However, the current BN techniques are mainly based on real-valued computation algorithms. To achieve CBN operations, the square root scaling method [27] is considered to perform the standard normal complex distribution (SNCD) processing for complex-valued SAR targets by multiplying the zero-centered data $(\mathbf{x} - \mathbb{E}[\mathbf{x}])$ and the reciprocal square root of a 2×2 covariance matrix $\mathbf{V}$, defined as

$$\hat{\mathbf{x}} = \mathbf{V}^{-\frac{1}{2}}(\mathbf{x} - \mathbb{E}[\mathbf{x}]) \tag{3}$$

where $\mathbf{x}$ indicates the complex-valued input, and the covariance matrix $\mathbf{V}$ is

$$\mathbf{V} = \begin{pmatrix} V_{rr} & V_{ri} \\ V_{ir} & V_{ii} \end{pmatrix} = \begin{pmatrix} \mathrm{Cov}(\Re(\mathbf{x}), \Re(\mathbf{x})) & \mathrm{Cov}(\Re(\mathbf{x}), \Im(\mathbf{x})) \\ \mathrm{Cov}(\Im(\mathbf{x}), \Re(\mathbf{x})) & \mathrm{Cov}(\Im(\mathbf{x}), \Im(\mathbf{x})) \end{pmatrix} \tag{4}$$

where Cov( ) represents the covariance calculation of different convolution input features. $\Re(\mathbf{x})$ and $\Im(\mathbf{x})$ symbolize the real and imaginary components of the complex-valued feature $\mathbf{x}$, respectively. The covariance matrix $\mathbf{V}$ is a positive definite (or semipositive definite) matrix, which ensures the existence of analytical solutions. After the above calculation, the complex-valued features obey the standard complex distribution of mean $\mu = 0$, covariance $\Gamma = 1$, and pseudocovariance $C = 0$. Accordingly, the formulation of CBN can be written as

$$\mathrm{CBN}(\hat{\mathbf{x}}) = \boldsymbol{\alpha} \cdot \hat{\mathbf{x}} + \beta \tag{5}$$

where $\boldsymbol{\alpha}$ and β represent the scaling factor and the bias constant, respectively, and both of them can be learned in the training phase. It is worth noting that β is a complex parameter with two learnable parameters, namely, the real part $\Re(\beta)$ and the imaginary part $\Im(\beta)$. The scaling factor $\boldsymbol{\alpha}$ is a 2×2 semipositive definite matrix, which is defined as

$$\boldsymbol{\alpha} = \begin{pmatrix} \alpha_{rr} & \alpha_{ri} \\ \alpha_{ir} & \alpha_{ii} \end{pmatrix} \tag{6}$$

where $\alpha_{ri} = \alpha_{ir}$. Overall, three learnable parameters in CBN need to be optimized in the process of network training. Here, we initialize α_{ri}, $\Re(\beta)$, and $\Im(\beta)$ to 0 and α_{rr} and α_{ii} to $1/\sqrt{2}$ to ensure that the variance of the real and imaginary components for the complex-valued SAR target feature is equal to 1.

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C. Complex-Valued Activation

The design of complex-valued activation needs to fully consider the correlation between the magnitude information and the phase information for SAR targets, instead of simply using real-valued activation functions, such as Sigmoid, ReLU, and so on. Here, we employ a complex-valued rectified linear unit (CReLU) activation function to consider the nonlinear activation of complex-valued SAR target features, by performing ReLU activation on the real and imaginary components, respectively

$$\mathrm{CReLU}(\mathbf{x}) = \mathrm{ReLU}(\Re(\mathbf{x})) + i\mathrm{ReLU}(\Im(\mathbf{x})). \tag{7}$$

D. Complex-Valued Pooling

Similar to the real-valued pooling layer, the implementation of complex-valued pooling mainly covers the following peculiarities: 1) downsampling the SAR target features; 2) removing redundant information of SAR target features by compressing feature information and reducing dimensionality; 3) realizing the nonlinear mapping of target information and improving the representation ability of the network through hierarchical structure mapping; and d) ensuring the invariance of SAR target features, such as translation, scale, and rotation invariance. In this section, we mainly introduce the two types of pooling methods: CMax-Pooling and CAvg-Pooling.

1) Complex-Valued Maximum Pooling (CMax-Pooling): Different from the real-valued Max-Pooling operation, the CMax-Pooling cannot simply select the maximum value of the real and imaginary parts in SAR target features. This operation will cause a "mismatch problem" between the magnitude information and the phase information. The main reason is that there is a strict one-to-one correspondence between the magnitude information and the phase information of the complex-valued SAR target. Any mismatch between magnitude and phase will make the SAR target information lose its physical meaning, thereby affecting the target recognition performance of the model. Therefore, in this article, the proposed CMax-Pooling method considers the matching relationship between magnitude information and phase information for the SAR target.

Assuming that the output of the SAR target after the activation of the $l+1$th layer is $O^{l+1}(\mathbf{x}) = \Re(\mathbf{x}) + i \cdot \Im(\mathbf{x})$, the Max-Pooling of the complex-valued magnitude is

$$\Pi^{l+1}(\mathbf{x}) = \underset{k,s,p}{\text{Max-Pooling}}\{|\mathbf{x}|\} \tag{8}$$

where $|\cdot|$ indicates the modulus calculation of complex-valued features. k, s, and p represent the pooling size, the step size, and the padding size, respectively. Next, find out the position index information in the original feature matrix

$$\mathcal{I}(\mathbf{x}) \leftarrow \mathrm{index}\left\{\Pi^{l+1}(\mathbf{x})\right\} \tag{9}$$

where index$\{\}$ is the location index of the feature maps after the target magnitude maximum pooling in the original feature map. After the Max-Pooling operation, the obtained features of the real and imaginary parts in the complex-valued SAR target are expressed as

$$\begin{cases} \hat{\Re}(\mathbf{x}) = \Re(\mathbf{x})\{\mathcal{I}(\mathbf{x})\} \\ \hat{\Im}(\mathbf{x}) = \Im(\mathbf{x})\{\mathcal{I}(\mathbf{x})\} \end{cases} \tag{10}$$

where $\hat{\Re}(\mathbf{x})$ and $\hat{\Im}(\mathbf{x})$ represent the real and imaginary information of $\Re(\mathbf{x})$ and $\Im(\mathbf{x})$ according to the position index information $\mathcal{I}(\mathbf{x})$, respectively. Eventually, the output of the CMax-Pooling is

$$\hat{O}^{l+1}(\mathbf{x}) = \hat{\Re}(\mathbf{x}) + i\hat{\Im}(\mathbf{x}). \tag{11}$$

2) Complex-Valued Average Pooling (CAvg-Pooling): The purpose of average pooling is to average the local features by means of slide window to achieve the dimensionality reduction and feature abstraction. For complex-valued SAR target features, under the premise of a fixed pooling size, the average pooling method will not cause the mismatch problem between magnitude and phase features. The main reason may be that each feature in the process of average pooling contributes to the results. Therefore, in this article, the CAvg-Pooling is achieved by averaging the real and imaginary parts of the SAR target separately

$$O^{l+1}(\mathbf{x}) = \text{Avg-Pooling}_{k,s,p}\{\Re(\mathbf{x})\} + i \cdot \text{Avg-Pooling}_{k,s,p}\{\Im(\mathbf{x})\}. \tag{12}$$

E. Complex-Valued Full Connection

The fully connected layer is widely applied in the DL network as a classifier due to its simplicity and effective feature extraction ability. Different from the real-valued fully connected layer, which performs a simple linear transformation of target features, the CFC needs to consider the combination of the real and imaginary parts in SAR target features. Suppose a given complex-valued SAR target feature $\mathbf{x} = \Re(\mathbf{x}) + i\Im(\mathbf{x}) = \mathbf{x}_r + i \cdot \mathbf{x}_i$, we have

$$\begin{cases} O_{rr} = \mathbf{w}_r \cdot \mathbf{x}_r + b_r \\ O_{ii} = \mathbf{w}_i \cdot \mathbf{x}_i \\ O_{ri} = \mathbf{w}_r \cdot \mathbf{x}_i \\ O_{ir} = \mathbf{w}_i \cdot \mathbf{x}_r + b_i \end{cases} \tag{13}$$

where $\mathbf{w}_r$ and $\mathbf{w}_i$ represent the weight of real and imaginary parts, respectively, and b_r and b_i denote the corresponding bias in the real and imaginary parts, respectively. By integrating (13), we can obtain the formulation of CFC as follows:

$$O_{\mathrm{CFC}} = (O_{rr} - O_{ii}) + i \cdot (O_{ri} + O_{ir}). \tag{14}$$

F. Complex-Valued Weight Initialization

In the training phase of DL networks, weight initialization can effectively accelerate the convergence speed of the network and avoid the problem of vanish gradient. This section follows the research work of [8], [25], and [27] to derive the initialization about complex-valued weight parameters.

Define a complex-valued weight variable $W = \Re(W) + i \cdot \Im(W) = |W|e^{i\theta}$, where $|W|$ and θ denote the magnitude and phase of the complex-valued weight, respectively. According to [27], we can obtain the variance of W

$$\mathrm{Var}(W) = \mathrm{Var}(|W|) + (\mathbb{E}[|W|])^2. \tag{15}$$

It can be concluded from (15) that the variance of W is related to its magnitude $|W|$, but not to its phase angle θ. Based

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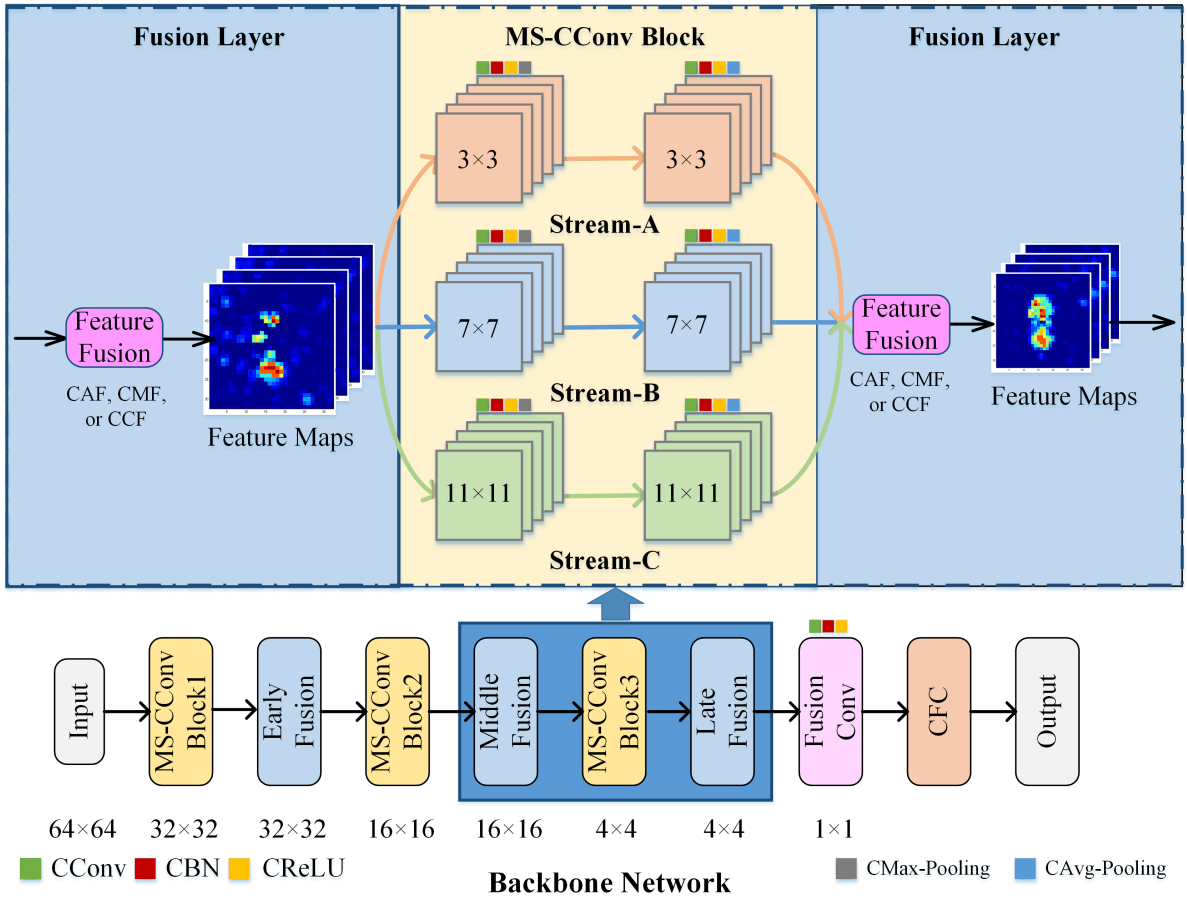

Fig. 2. Architecture of the proposed MS-CVNets.

on the Rayleigh distribution, the expectation and variance of the single variable are expressed as

$$\mathbb{E}[|W|] = \sqrt{\frac{\pi}{2}} \cdot \sigma \tag{16}$$

$$\mathrm{Var}(|W|) = \frac{4-\pi}{2} \cdot \sigma^2 \tag{17}$$

where σ indicates the scale parameter, which is used to control the width of the Rayleigh distribution envelope. Therefore, we can deduce the variance of W as

$$\mathrm{Var}(W) = \frac{4-\pi}{2} \cdot \sigma^2 + \left(\sqrt{\frac{\pi}{2}} \cdot \sigma\right)^2 = 2\sigma^2. \tag{18}$$

According to (18), it can be clearly seen that the complex-valued weight W is positively correlated with the scaling parameter σ. In this article, we adopt the method of He *et al.* [8] to initialize the weight parameter W by the following operation:

$$\mathrm{Var}(W) = 2/n_{\mathrm{in}}, \sigma = 1/\sqrt{n_{\mathrm{in}}} \tag{19}$$

where n_{in} is the number of the input channel. Finally, according to the mutual conversion of polar coordinates and rectangular coordinates, the real and imaginary parts after complex-valued weight initializations are expressed as

$$\begin{cases} \Re(W) = |W| \cdot \cos(\theta) \\ \Im(W) = |W| \cdot \sin(\theta) \end{cases} \tag{20}$$

where θ denotes the phase angle, initialized to the uniform distribution with $[-\pi/2, +\pi/2]$ in the training phase.

III. Multi-Stream Complex-Valued Networks

In this section, we formulate the methodology of the MS-CVNets systematically, and the architecture of MS-CVNets is shown in Fig. 2. The basic blocks of MS-CVNets adopt the MS CConv block (MS-CConv block) and the fusion layer. Specifically, an MS-CConv block involves three CConv operation flows in parallel, each CConv flow is composed of two cascaded convolutional layers, and each convolutional layer follows the CConv, CBN, CReLU, and CMax-Pooling processing in an orderly manner. Fig. 3 displays the parameter configuration of an MS-CConv block. It should be noted that the architecture of MS-CConv Block1 and MS-CConv Block2 is strictly consistent with Fig. 3, while the first convolutional layer of MS-CConv Block3 is CMax-Pooling, and the second convolutional layer, CAvg-Pooling instead, as shown in Fig. 2.

In the training process, the input complex-valued SAR target first passes through the MS-CConv block to perform feature extraction and dimensionality reduction for SAR targets. Then, the extracted MS target features are fused and represented by different feature fusion algorithms. By applying the fusion convolution layer and the fully connection, the final feature

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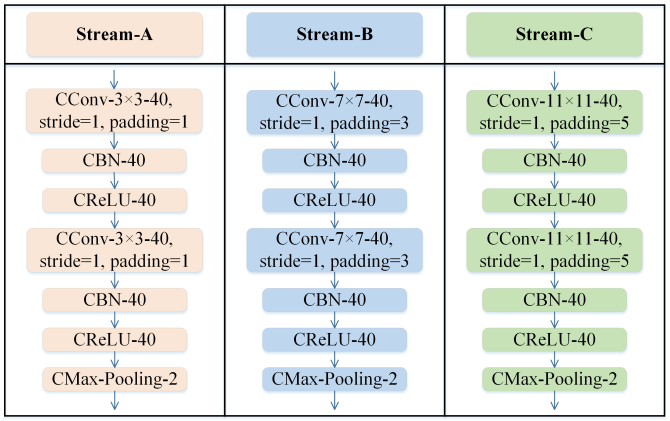

Fig. 3. Composition of MS-CConv block.

vector is obtained to achieve the classification and recognition of SAR targets.

A. Multi-Stream Structure

In SAR-ATR, high-precision target identification in an interference environment has always been a challenging task, owing to the effect of noise interference, target attitude changes, and version variation (VV). In this section, we introduce the MS structure [38]–[40] to the task of SAR target recognition, by leveraging different convolution kernels to extract significant multiscale complex-valued target features. Here, the kernel sizes of 3×3, 7×7, and 11×11 in the convolution operation are utilized to acquire MS features of complex-valued SAR targets, corresponding to Stream-A, Stream-B, and Stream-C, as shown in Fig. 3. Furthermore, to maintain the consistency of the feature streams in different kernel sizes, we set the 40-channel feature for each stream and employ a unique feature fusion algorithm to make full use of the extracted multiscale feature information.

In general, the larger the size of the convolution kernel, the larger the receptive field of its features. Therefore, the MS-CConv processing can generate a wealth of multiscale feature information of SAR targets, which improves the recognition performance of the model. In addition, the MS structure has the advantages of alleviating the vanishing gradient and enhancing the generalization ability of the model, which is in line with the aforementioned SAR-ATR task and provides us with a solution.

B. Complex-Valued Feature Fusion

After the MS structure processing, a wealth of multiscale target feature information can be effectively extracted. Then, how to fuse and represent these target features is crucial to the subsequent targets recognition. To this end, this article proposes three types of feature fusion strategies to characterize and integrate MS feature information of SAR targets, containing CAF, CMF, and CCF.

1) Complex Average Fusion (CAF): That is, adopting average calculation to perform fusion representation of MS SAR target features. The obtained mean result can effectively retain the background, block area, and local feature information of MS features, so that each stream will produce a certain degree of contribution to the subsequent fused feature. Assuming that the MS complex-valued features are $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \ldots, \mathbf{F}_N$, respectively, the CAF can be described as

$$\mathbf{F} = \frac{1}{N}\sum_{n=1}^{N}(\mathbf{F}_n) = \frac{1}{N}\sum_{n=1}^{N}(\Re(\mathbf{F}_n)) + i \cdot \frac{1}{N}\sum_{n=1}^{N}(\Im(\mathbf{F}_n)) \quad (21)$$

where N denotes the number of feature streams.

2) Complex Maximum Fusion (CMF): The bright points, lines, and regions of the feature map often contain useful information about the target. This information plays a valuable role in the subsequent target feature extraction and target classification task. Besides, consistent with the abovementioned CMax-Pooling operation, we propose a CMF fusion method to avoid the mismatch of magnitude and phase information in complex-valued features.

The corresponding elementwised maximum magnitude feature matrix $|\mathbf{F}|$ is denoted as

$$|\mathbf{F}| = \max(|\mathbf{F}_1|, |\mathbf{F}_2|, |\mathbf{F}_3|, \ldots, |\mathbf{F}_N|) \quad (22)$$

where max( ) indicates the calculation of the maximum value of the corresponding element in the different magnitude matrix. According to the position index information corresponding to the maximum value of $|\mathbf{F}|$, represented by

$$\mathcal{I}(|\mathbf{F}|) \leftarrow \text{index}\{|\mathbf{F}|\}. \quad (23)$$

We can obtain the real and imaginary parts of the complex-valued SAR target feature as

$$\begin{cases} \hat{\Re}(\mathbf{F}) = \Re(\mathbf{F})\{\mathcal{I}(|\mathbf{F}|)\} \\ \hat{\Im}(\mathbf{F}) = \Im(\mathbf{F})\{\mathcal{I}(|\mathbf{F}|)\}. \end{cases} \quad (24)$$

Finally, the extracted feature matrix after the CMF processing is expressed as

$$\hat{\mathbf{F}} = \hat{\Re}(\mathbf{F}) + i \cdot \hat{\Im}(\mathbf{F}). \quad (25)$$

CMF can effectively retain the high-magnitude information in the MS complex-valued features, characterized as the edge and profile information of the target, thereby improving the representation ability of MS-CVNets and providing guarantee for subsequent target recognition.

3) Complex Concatenation Fusion (CCF): In the process of feature fusion, a more intuitive and effective method is to directly concatenate the feature information of multiple data streams. Inspired by the research work in [41] and [42], this article proposes a CCF fusion method into the complex-valued SAR target recognition, and its implementation principle is shown in Fig. 4. The multiscaled features' information extracted by the previous MS-CConv block is first concatenated into the real and imaginary parts of new complex-valued features. Here, the number of channels is multiplied, from N to $3N$. Then, the concatenated features are input into the next MS-CConv block for feature fusion and extraction, constituting the real and imaginary parts of a new feature stream. For a single feature stream, this is a feature compression and sorting process, and the number of feature channels decreases from $3N$ to N. The CCF processing

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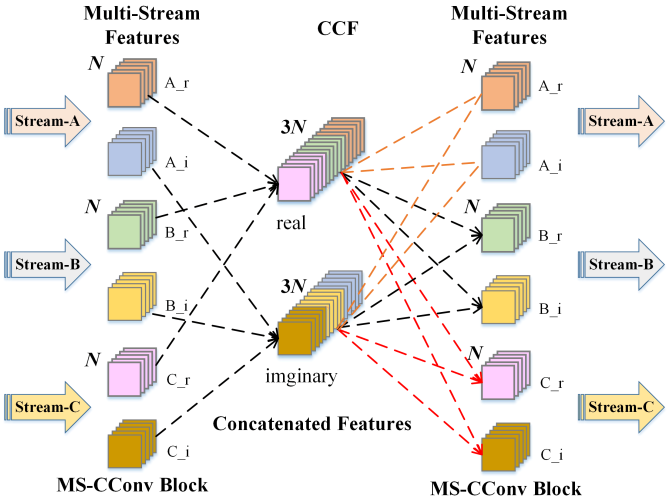

Fig. 4. Diagram of CCF feature fusion.

is similar to the encoding process in the autoencoder, which can effectively remove redundant information of features and extract more universal target features.

Meanwhile, based on the abovementioned three fusion strategies, MS-CVNets leverage a unique MS complex-valued triple fusion design to further improve the representation ability of the model, namely, early fusion, middle fusion, and late fusion, as shown in Fig. 2.

C. Parameter Update

This section mainly introduces the principle of loss back-propagation in the proposed MS-CVNets. Since the category labels of SAR targets are real-valued, we employ the modular calculation on the real and imaginary parts of the output of fully connection layer during the training process

$$|\mathbf{Y}| = \left(\Re(\mathbf{Y})^2 + \Im(\mathbf{Y})^2\right)^{1/2} \tag{26}$$

where $\mathbf{Y}$is the complex-valued output of fully connection layer. Suppose the training category label is T, the loss function of MS-CVNets based on the cross entropy (CE) measurement is defined as

$$\mathcal{L} = D_{KL}(|\mathbf{Y}| \parallel \mathbf{T}) = \sum_{i=1}^{m} |\mathbf{Y}_i| \cdot \log \frac{|\mathbf{Y}_i|}{\mathbf{T}_i} \tag{27}$$

where m represents the number of target samples, and $\mathbf{Y}_i$ and $\mathbf{T}_i$ indicate the ith output and its corresponding category label, respectively. Assume that the output of the node O_i^{l+1} at the $l+1$th layer is

$$\begin{cases} V_i^{l+1} = \sum_{j=1}^{K} w_{ij}^{l+1} \cdot O_j^l + b_i^{l+1} \\ O_i^{l+1} = \mathrm{ReLU}\left(\Re\left(V_i^{l+1}\right)\right) + i \cdot \mathrm{ReLU}\left(\Im\left(V_i^{l+1}\right)\right) \end{cases} \tag{28}$$

where K denotes the number of node at layer l. w_{ij}^{l+1} and b_i^{l+1} are the weight and bias between the node O_i^{l+1} of the $l+1$th layer and the node O_j^l of the lth layer, respectively. The stochastic gradient descent (SGD) algorithm is selected to optimize the loss function in (27), and the updating of hyperparameters is expressed as

$$w_{ij}^{l+1}(t+1) = w_{ij}^{l+1}(t) + \Delta w_{ij}^{l+1}(t) = w_{ij}^{l+1}(t) - \eta \frac{\partial \mathcal{L}}{\partial w_{ij}^{l+1}(t)} \tag{29}$$

$$b_i^{l+1}(t+1) = b_i^{l+1}(t) + \Delta b_i^{l+1}(t) = b_i^{l+1}(t) - \eta \frac{\partial \mathcal{L}}{\partial b_i^{l+1}(t)} \tag{30}$$

where $w_{ij}^{l+1}(t)$ and $b_i^{l+1}(t)$ indicate the weight and bias of the tth iteration update, respectively, and η is the learning rate of the SGD optimizer. According to the chain derivation rule, we can obtain the following partial derivative formula:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w_{ij}^{l+1}} &= \frac{\partial \mathcal{L}}{\partial \Re\left(w_{ij}^{l+1}\right)} + \frac{\partial \mathcal{L}}{\partial \Im\left(w_{ij}^{l+1}\right)} \\ &= \left(\frac{\partial \mathcal{L}}{\partial \Re\left(V_i^{l+1}\right)} \frac{\partial \Re\left(V_i^{l+1}\right)}{\partial \Re\left(w_{ij}^{l+1}\right)} + \frac{\partial \mathcal{L}}{\partial \Im\left(V_i^{l+1}\right)} \frac{\partial \Im\left(V_i^{l+1}\right)}{\partial \Re\left(w_{ij}^{l+1}\right)}\right) \\ &\quad + i\left(\frac{\partial \mathcal{L}}{\partial \Im\left(V_i^{l+1}\right)} \frac{\partial \Im\left(V_i^{l+1}\right)}{\partial \Im\left(w_{ij}^{l+1}\right)} + \frac{\partial \mathcal{L}}{\partial \Re\left(V_i^{l+1}\right)} \frac{\partial \Re\left(V_i^{l+1}\right)}{\partial \Im\left(w_{ij}^{l+1}\right)}\right) \end{aligned} \tag{31}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial b_i^{l+1}} &= \frac{\partial \mathcal{L}}{\partial \Re(b_i^{l+1})} + \frac{\partial \mathcal{L}}{\partial \Im(b_i^{l+1})} \\ &= \left(\frac{\partial \mathcal{L}}{\partial \Re\left(V_i^{l+1}\right)} \frac{\partial \Re\left(V_i^{l+1}\right)}{\partial \Re\left(b_i^{l+1}\right)} + \frac{\partial \mathcal{L}}{\partial \Im\left(V_i^{l+1}\right)} \frac{\partial \Im\left(V_i^{l+1}\right)}{\partial \Re\left(b_i^{l+1}\right)}\right) \\ &\quad + i\left(\frac{\partial \mathcal{L}}{\partial \Im\left(V_i^{l+1}\right)} \frac{\partial \Im\left(V_i^{l+1}\right)}{\partial \Im\left(b_i^{l+1}\right)} + \frac{\partial \mathcal{L}}{\partial \Re\left(V_i^{l+1}\right)} \frac{\partial \Re\left(V_i^{l+1}\right)}{\partial \Im\left(b_i^{l+1}\right)}\right). \end{aligned} \tag{32}$$

By substituting the above partial derivative results into (29) and (30), the update of complex-valued parameters in MS-CVNets is accomplished.

As a high precision and robust SAR target recognition model, the performance of MS-CVNets is primarily guaranteed by the following two aspects: 1) the complex-valued model. Through the introduction of complex-valued calculation modules, the phase information in the complex-valued SAR target is effectively utilized to enhance the recognition ability of target features and 2) complex-valued MS feature fusion. The MS structure produces a wealth of multiscale target feature information, which not only considers a more complete feature extraction system but also effectively alleviates the vanishing gradient in the training phase. From the perspective of complex-valued information processing, MS-CVNet is a target recognition framework tailored for SAR-ATR. However, just like a coin has two sides, while MS-CVNets possess excellent SAR target recognition performance, it bears some disadvantages, such as complex model architecture and heavy computation burden.

IV. Experimental Results Analysis and Discussion

This section mainly conducts the experimental analysis and discussion on the proposed MS-CVNets. In addition, four classic and state-of-the-art recognition approaches related to the SAR-ATR task are selected for comparison, including

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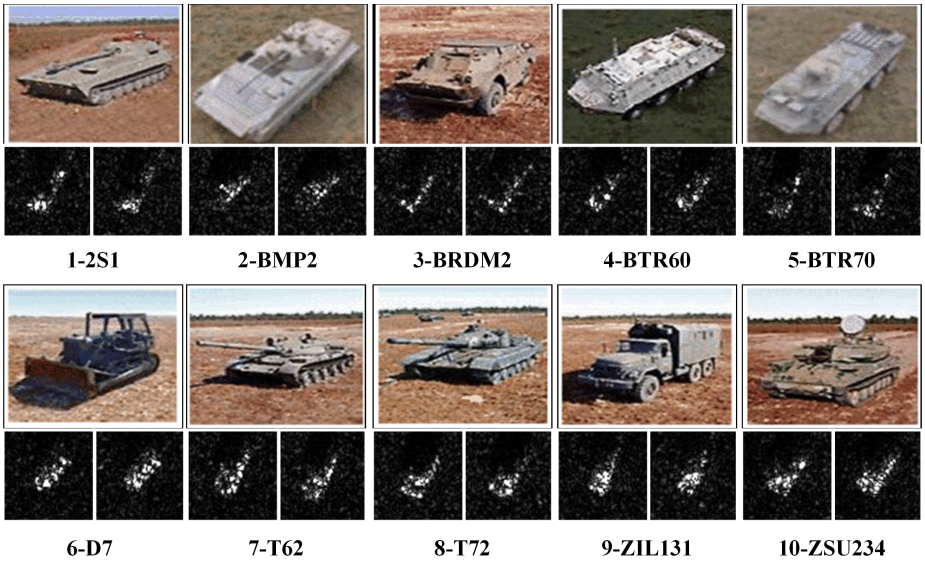

Fig. 5. Optical-complex SAR image comparison. (Top) Optical image. (Bottom left) Real part. (Bottom right) Imaginary part.

the Traditional-CNN [7], A-ConvNet [14], FEN [17], and CV-CNN [28]. In our case, the proposed MS-CVNet is implemented on the Pytorch 1.6 platform with Intel Core i7-5930K CPU and Nvidia GeForce GTX 1080Ti GPU.

In this article, we adopt accuracy and CE loss (CEL) as evaluation metrics, and the definition of accuracy can be written as

$$Acc = \frac{\text{Number of correct targets}}{\text{Number of total targets}}. \tag{33}$$

The CEL metric is defined in (27), and for the sake of fairness, all comparative experiments in this article employ this evaluation criterion as the loss indicator.

A. Production and Preparation of the Complex-Valued SAR Target Dataset

The MSTAR dataset[1] [14], [17], [44], supported by the Defense Advanced Research Projects Agency (DARPA), X-band with 0.3 m $\times$ 0.3 m resolution, HH polarization, and 360° articulation with a 1° spacing, is widely applied in the field of SAR-ATR. However, to the best of our knowledge, most of the current target recognition methods [14]–[17] adopting this dataset only use the magnitude information of the SAR target and ignore its phase information. To effectively exploit the phase information of the SAR target, we organize a complex-valued SAR target dataset based on the original MSTAR dataset, which contains the real and imaginary parts of the SAR target[2]. In the production process, the region of interest with a size of 64 $\times$ 64 is selected for SAR sample generation in the MSTAR dataset. The target types, quantities, serial numbers, and depression angles of the MSTAR dataset are listed in Table I, and the corresponding optical-complex SAR image is depicted in Fig. 5. Moreover, according to the research work of Kechagias-Stamatis *et al.* [34]–[37], [43], we divide the MSTAR dataset with complex-valued information into standard operating condition (SOC) dataset and extended operating condition (EOC) dataset in order to facilitate subsequent MS-CVNets performance testing.

B. SOC Experiments

The SOC experiments mainly include the ten types and the three types of target configuration tests. Before the SOC test, it is necessary to configure the corresponding SOC-10 and SOC-3 target datasets. The SOC-10 dataset contains all ten types of targets, while the SOC-3 dataset only contains three types of targets 2-BMP2, 5-BTR70, and 8-T72. The specific configurations are shown in Table II.

1) Ten-Type Test: The SOC-10 target test is a comprehensive test of all target types based on the configuration of the SOC-10 dataset. It can be seen from Table II that the train set has 2747 samples with a depression angle of 17°, and the test set has 2425 samples with a depression angle of 15°. Owing to the diversity and completeness of the SAR targets, the comprehensive performance of the model for all targets can be effectively reflected through SOC-10 testing.

Table III reports the quantitative test results on the SOC-10 dataset. It can be observed from Table III that the accuracy results of five SAR target recognition models are ranked from high to low, namely, MS-CVNtes-99.79%, FEN-99.63%, A-ConvNet-99.13%, CV-CNN-98.10%, and traditional-CNN-94.72%. The proposed MS-CVNets have achieved superior results in the accuracy of the SOC-10 dataset, surpassing the other four SAR target recognition models. In addition, in regard to the test loss, it can reveal the fitting degree of the model to the target sample: the loss of MS-CVNets is notably lower than that of other four models. For illustrative purposes, the confusion matrix of MS-CVNets on the SOC-10 dataset is displayed in Fig. 6. It can be seen that MS-CVNets has achieved 100% recognition accuracy on seven types of targets including BMP2, BRDM2, BTR60, BTR70, T62, ZIL131, and

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TABLE I
MSTAR DATASET INFORMATION

Target	Serial No.	Train 17°	Test 15°	Test 30°	Test 45°
1-2S1	b01	299	274	288	288
2-BMP2	9563	233	195	-	-
	9566	232	196	-	-
	c21	233	196	-	-
3-BRDM2	e71	298	274	288	288
4-BTR60	k10yt7532	256	195	-	-
5-BTR70	c71	233	196	-	-
6-D7	-	299	274	-	-
7-T62	a51	299	273	-	-
8-T72	132	232	196	-	-
	812	231	195	-	-
	s7	228	191	-	-
9-ZIL131	e21	299	274	-	-
10-ZSU234	d08	299	274	288	303

TABLE II
EOC AND SOC SUBDATASETS CONFIGURATION

Datasets	Training Targets	Testing Targets	Description
SOC-10	1-2S1, 2-BMP2 (9563), 3-BRDM2, 4-BTR60, 5-BTR70, 6-D7, 7-T62, 8-T72 (132), 9-ZIL131, 10-ZSU234		Both for training and testing
SOC-3	2-BMP2 (9563), 2-BMP2 (9566), 2-BMP2 (c21), 5-BTR70, 8-T72 (132), 8-T72 (812), 8-T72 (s7)		Both for training and testing
EOC-DV-3	1-2S1, 3-BRDM2, 10-ZSU234	1-2S1, 3-BRDM2, 10-ZSU234	Training at 17° depression angle. Testing at 15°, 30°, 45° depression angles.
EOC-VV-4	2-BMP2 (9653), 4-BTR60, 7-T62, 8-T72(132)	2-BMP2 (9566), 2-BMP2 (c21), 4-BTR60, 7-T62, 8-T72(812), 8-T72 (s7)	Training at 17° depression angle. Testing at 15° against BMP2 and T72 version variants.
EOC-NV-10	1-2S1, 2-BMP2 (9563), 3-BRDM2, 4-BTR60, 5-BTR70, 6-D7, 7-T62, 8-T72 (132), 9-ZIL131, 10-ZSU234		Training at 17° depression angle. Testing at 15° with additive white Gaussian noise at SNR = {-10, -5, 0, 5, 10} dB.

TABLE III
QUANTITATIVE TEST RESULTS ON THE SOC-10 DATASET. THE BEST ONE IS SHOWN IN BOLD

Datasets	Models	Model Type	Loss	ACC (%)
SOC-10	Traditional-CNN	real-valued	4.0950	94.72
	A-ConvNet	real-valued	1.1424	99.13
	FEN	real-valued	0.5264	99.63
	CV-CNN	complex-valued	2.0521	98.10
	MS-CVNets	complex-valued	**0.2848**	**99.79**

ZSU234, and the recognition accuracies of targets 2S1, D7, and T72 are 99.3%, 99.3%, and 99.5%, respectively.

2) Three-Type Test: The SOC-3 target experiment is to identify the three types of targets 2-BMP2, 5-BTR70, and 8-T72. The specific configuration of the SOC-3 dataset is exhibited in Table II. The dataset contains three types and seven series of target samples. Specifically, the train set and the test set have 1622 and 1365 target samples, respectively. Through the test of the SOC-3 dataset, it can clearly reflect the learning and representation ability of the model for specific targets.

The quantitative test results on the SOC-3 dataset and the corresponding confusion matrix are listed in Table IV and Fig. 7. It can be seen from the SOC-3 experimental results that all the five target recognition approaches have achieved high recognition accuracy. From high to low, they are

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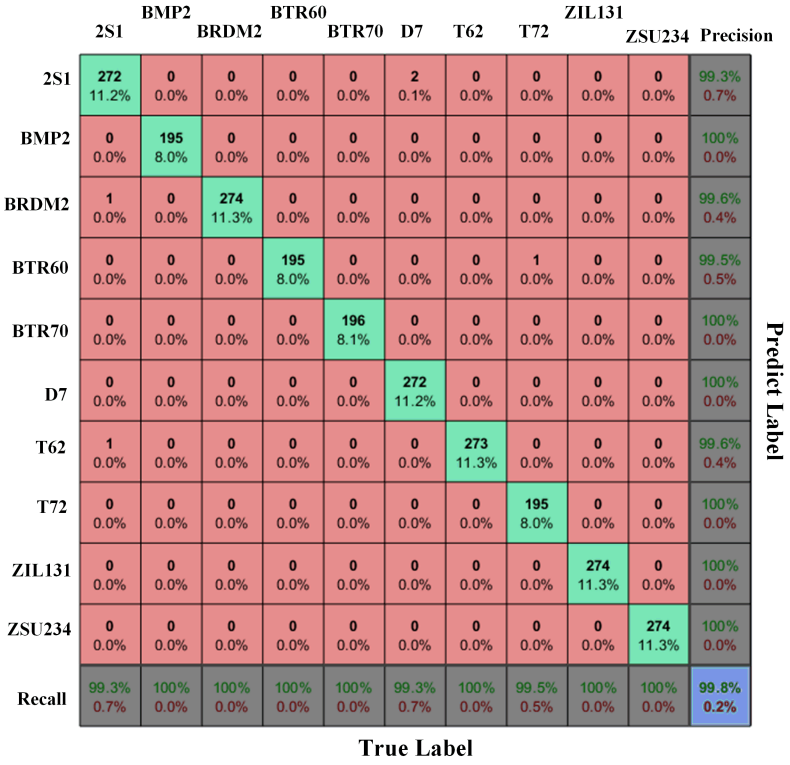

Fig. 6. Confusion matrix of MS-CVNets on the SOC-10 dataset.

TABLE IV
QUANTITATIVE TEST RESULTS ON THE SOC-3 DATASET. THE BEST ONE IS SHOWN IN BOLD

Datasets	Models	Model Type	Loss	ACC (%)
SOC-3	Traditional-CNN	real-valued	0.1132	99.71
	A-ConvNet	real-valued	0.0928	99.85
	FEN	real-valued	0.0746	99.93
	CV-CNN	complex-valued	0.2488	99.56
	MS-CVNets	complex-valued	**0.0251**	**100.00**

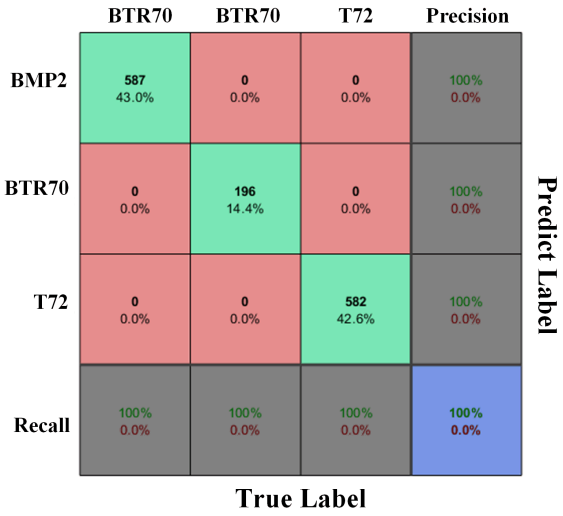

Fig. 7. Confusion matrix of MS-CVNets on the SOC-3.

MS-CVNets (100%), FEN (99.93%), A-ConvNet (99.85%), traditional-CNN (99.71%), and CV-CNN (99.56%). In terms of loss error, MS-CVNets have the lowest test error of 0.0251, while CV-CNN has the highest test error, namely, 0.2488. Overall, the proposed MS-CVNets show stronger performance than the existing target recognition methods in both of target recognition accuracy and target fitting error, further illustrating that the introduction of phase information has a positive effect on the improvement of target recognition performance. Moreover, the MS structure also promotes MS-CVNets to learn more universal features of SAR targets, which improves the generalization ability of the model. In summary, based on the experimental results on SOC-10 and SOC-3 datasets, MS-CVNets can obtain superior SAR target recognition performance compared with the current state-of-the-art real-valued model, validating its effectiveness for the SAR-ATR task in real scenarios.

C. EOC Experiments

Different from the SOC dataset that only considers target samples with two type depression angles (training 17° and testing 15°), the EOC datasets contain target samples under the condition of various depression angles (depression angle

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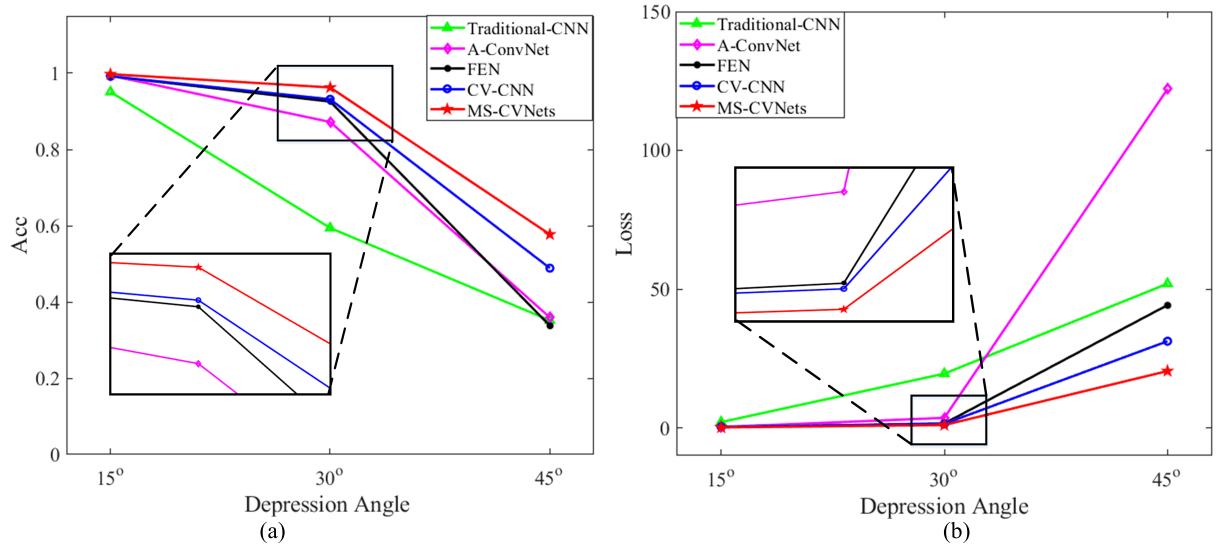

Fig. 8. EOC-DV-3 dataset accuracy and loss curves. (a) Target accuracy versus depression angle. (b) Test loss versus depression angle.

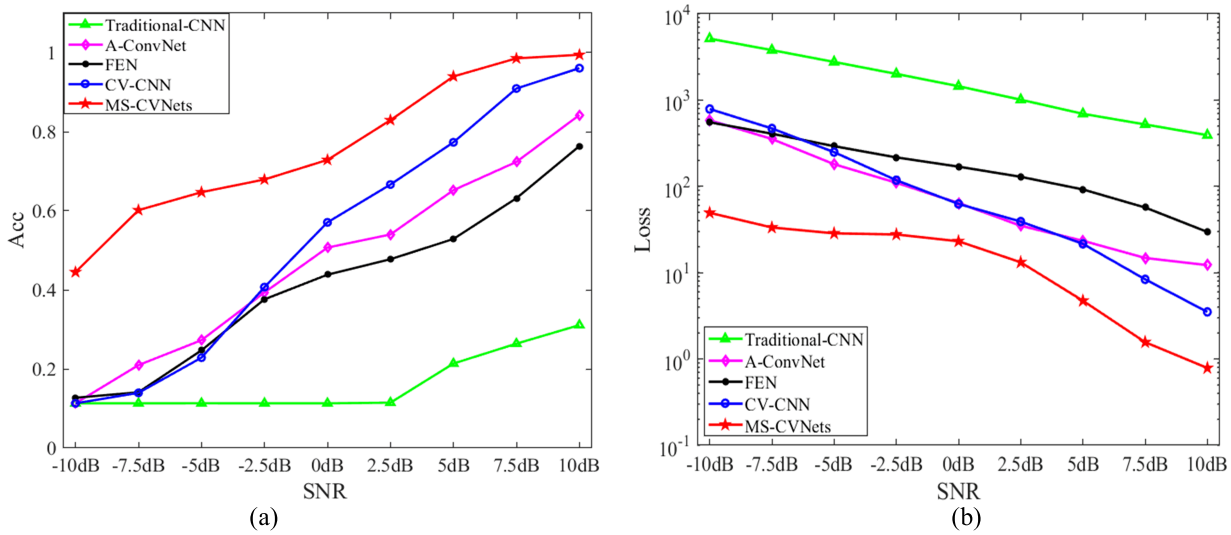

Fig. 9. EOC-NV-10 dataset accuracy and loss curve. (a) Target accuracy versus SNR variation curve. (b) Test loss versus SNR variation curve (logarithm calculation of numerical).

from 15° to 45°), the state change (version, configuration, and shielding), and the environmental noise change. Therefore, this section conducts the experimental tests of MS-CVNets under these three EOCs, namely, depression variation (DV) test, VV test, and noise variation (NV) test, and the corresponding dataset configurations are listed in Table II.

1) Depression Variation: The DV experiment aims to test the recognition performance of the model with target samples obtained under different depression angles. The configuration of the DV dataset is listed in Table II, EOC-DV-3. Specifically, the train set contains target samples with a depression angle of 17°, and the test set utilizes target samples with the depression angles of 15°, 30°, and 45°, respectively.

The test results of the EOC-DV-3 dataset are displayed in Table V and Fig. 8. It can be seen that as the depression angle gradually increases (from 15° to 45°), the target recognition performance of all models gradually declines. Particularly, in the 15°–30° stage, the target recognition performance of the FEN, CV-CNN, and MS-CVNets drops more slowly than that of the Traditional-CNN and A-ConvNet. However, in the 30°–45° stage, the target accuracies of all the five models decrease dramatically. Comparing with other state-of-the-art approaches, the proposed MS-CVNets can achieve less performance degradation, indicating that the model based on complex-value calculation has stronger ability to resist more depression angle variations than the model based on real-value calculation. Based on the abovementioned experimental analysis, we can conclude that the proposed MS-CVNets achieves stronger target recognition performance than other models when the depression angle is 15°, 30°, and 45°, illustrating strong scene change robustness of MS-CVNet when facing the large variation of target depression angle.

2) Version Variation: In the SAR-ATR application, as the battlefield environment and mission requirements change rapidly, the existence of VV poses a challenge for SAR target recognition. In response to this problem, this section employs the EOC-VV-4 dataset to test the influence of VV on different target recognition models. Table II displays the

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TABLE V
QUANTITATIVE TEST RESULTS ON THE EOC-DV-3 DATASET. THE BEST ONE IS SHOWN IN BOLD

Datasets	Models	Depression 15°		Depression 30°		Depression 45°	
		Loss	Acc (%)	Loss	Acc (%)	Loss	Acc (%)
EOC-DV-3	Traditional-CNN	2.0609	95.09	19.5114	59.41	51.9380	35.16
	A-ConvNet	0.2673	99.39	3.5794	87.21	122.0737	35.93
	FEN	0.2279	99.19	1.5623	92.55	44.2105	33.77
	CV-CNN	0.3963	99.19	1.4383	93.17	31.1941	48.87
	MS-CVNets	**0.1313**	**99.76**	**0.9876**	**96.26**	**20.4512**	**57.72**

TABLE VI
QUANTITATIVE TEST RESULTS ON THE EOC-VV-4 EXPERIMENTAL RESULTS. THE BEST ONE IS SHOWN IN BOLD

Datasets	Models	Model Type	Loss	Acc (%)
EOC-VV-4	Traditional-CNN	real-valued	3.7074	88.84
	A-ConvNet	real-valued	3.3033	91.73
	FEN	real-valued	2.2924	93.42
	CV-CNN	complex-valued	4.3887	91.17
	MS-CVNets	complex-valued	**1.3103**	**96.71**

TABLE VII
QUANTITATIVE TEST RESULTS ON THE EOC-NV-10 EXPERIMENTAL RESULTS. THE BEST ONE IS SHOWN IN BOLD

Datasets	**Models**	**SNR (depression 15°)**				
		-10dB	**-5dB**	**0dB**	**5dB**	**10dB**
EOC-NV-10	Traditional-CNN	11.30%	11.30%	11.30%	21.36%	31.09%
	A-ConvNet	11.30%	27.27%	50.69%	65.21%	84.23%
	FEN	12.70%	24.70%	43.84%	52.87%	76.37%
	CV-CNN	11.30%	22.85%	57.07%	77.32%	96.08%
	MS-CVNets	**44.49%**	**64.70%**	**72.87%**	**93.98%**	**99.46%**

configuration of the EOC-VV-4 dataset, including four types of training target samples with a depression angle of 17° and the corresponding test target samples with a depression angle of 15°. Note that the targets of two-BMP2 and eight-T72 have different serial numbers on the EOC-VV-4 dataset.

Table VI lists the quantitative results of the VV experiment. The complex-valued calculation models, such as MS-CVNets and CV-CNN, fail to reveal significantly better target recognition performance than the real-valued calculation models, but MS-CVNets offer higher target recognition accuracy than other comparison models. In terms of loss, MS-CVNets also achieves the lowest test loss, namely, 1.3103, demonstrating its stronger generalization ability to VVs of SAR targets.

3) Noise Variation: The NV experiment aims to test the model's anti-interference ability under different intensities of noise. The better the anti-interference ability, the stronger the adaptability of the model to environmental changes. Here, the NV dataset, called EOC-NV-10, is built in this section, and the corresponding data configuration is shown in Table II. The train set samples mainly exploit the train samples of the SOC-10 dataset, and the test set is the test samples of the SOC-10 dataset with different levels of additive white Gaussian noise (AWGN) ranging from −10 to 10 dB signal-to-noise ratio (SNR).

The NV experimental results are listed in Table VII. Obviously, under the condition of high SNR (10 dB), MS-CVNets rank first with an accuracy rate of 99.46%; CV-CNN follows closely with an accuracy rate of 96.08%; A-ConvNet and FEN list third and four, respectively, while Traditional-CNN acquires the lowest accuracy rate, 31.09%. As the SNR gradually decreases, the target recognition accuracies of all the five models also gradually decline. Specifically, the recognition rates of FEN and Traditional-CNN models drop vastly and eventually diminish to 11% around, indicating that these two models are unable to train and learn in this case. On the contrary, models based on complex-valued calculations, such as CV-CNN and MS-CVNets, have achieved higher recognition accuracy than real-valued models (e.g., Traditional-CNN, A-ConvNet, and FEN). Fig. 9 presents the curve of the test accuracy and loss relationship with the SNR. It is clear that the proposed MS-CVNets can obtain more accurate recognition performance and lowest loss results, demonstrating the noise robustness of the proposed MS-CVNets.

D. Ablation Experiments

To validate the essentiality of the proposed MS-CVNets model, a series of ablation experiments on different network modules are carried out in this section, including MS structure, CAvg-Pooling, CMax-Pooling, and complex-valued feature fusion analysis.

1) Multi-Stream Structure: The MS structure ablation experiment mainly compares the performance test of MS-CVNets and the other models on the SOC-10 and EOC-NV-10 datasets. Here, MS-CVNets employs three types of convolution kernels (e.g., 3, 7, and 11) in the MS structure to achieve the multiscale feature extraction. Four single-stream complex-valued convolutional SAR target recognition models with convolution kernel sizes of 3, 7, and 11 are established for

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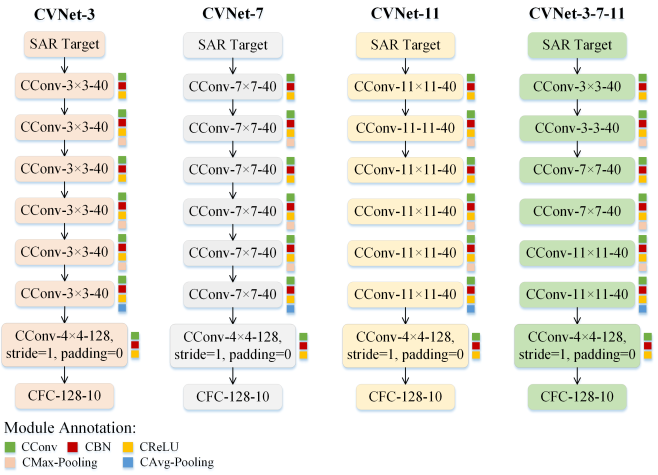

Fig. 10. Architecture of CVNet-3, CVNet-7, CVNet-11, and CVNet-3-7-11.

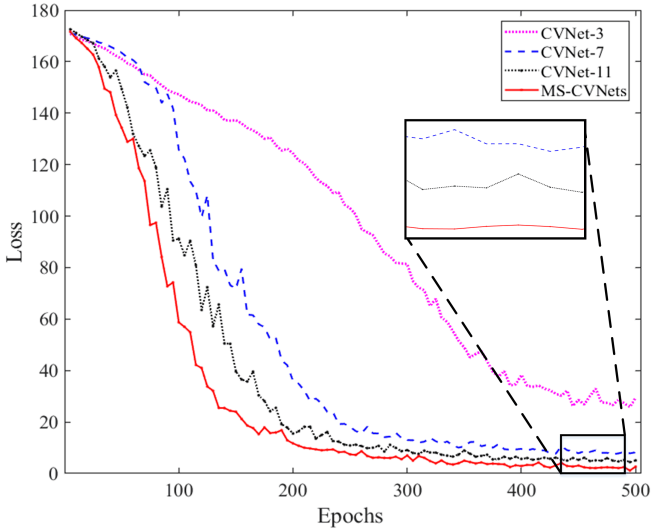

Fig. 11. Test loss curve of MS-CVNets, CVNet-3, CVNet-7, and CVNet-11.

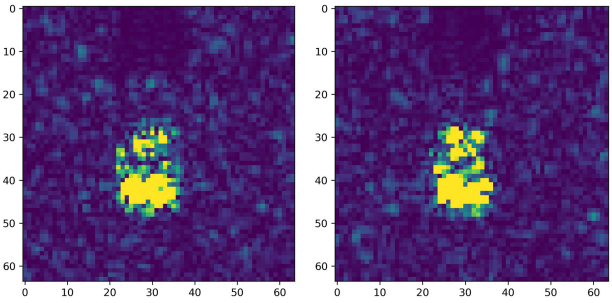

Fig. 12. Complex SAR target-BMP2 original image. (a) Real part. (b) Imaginary part.

comparison, respectively, referred to as CVNet-3, CVNet-7, CVNet-11, and CVNet-3-7-11, and the corresponding network architecture is shown in Fig. 10. In addition, in order to the effectiveness of the phase information, we build a real-valued network, MS-RVNet, which is identical with the MS-CVNets architecture.

Table VIII reports the ablation analysis of the proposed MS structure. As indicated from the experimental results, our proposed MS-CVNets model can achieve higher SAR target recognition accuracy and lower test loss than the single-stream models on both SOC-10 and EOC-NV-10 datasets. It is worth noting that the target test performance of CVNet-3-7-11 drops sharply on the EOC-NV-10 dataset, which shows that it is not robust to noise. The underlying reason may be the perception difference between different convolution kernels, and the fusion of convolution kernels of multiple sizes on a single-stream model will deepen this difference and degrade the generalization performance of the model. Fig. 11 shows the test loss curves of different target recognition models on the SOC-10 dataset. Moreover, MS-CVNets also achieve better target recognition performance than the real-valued model of the same architecture, MS-RVNet (Acc-99.51% on SOC-10 and ACC-86.68% on EOC-NV-10). Therefore, it can be concluded that the MS structure can effectively improve the representation ability for SAR target features and MS-CVNets have better target recognition performance than real-valued models of the same architecture.

Fig. 12 displays the real and imaginary images of the complex-valued SAR target-BMP2. Visually, both the real and imaginary images contain some similar feature information, such as the profile, shape, size, and detail texture of the target, but each image has its own emphasis on feature details owing to the introduction of the target phase information. Fig. 13 visualizes the intermediate feature maps of the BMP2 target after being processed by the first MS-CConv block. Specifically, Fig. 13(a)–(c) indicates the real and imaginary parts of the complex-valued targets after the Streams-A, B, and C convolution calculations, respectively. It can be seen that as the number of convolutional layer increases, the target features gradually convert from local detailed texture features to more abstract outlines, shadows, and spatial features. In addition, from the perspective of the convolution kernel size, the larger the size of the convolution kernel, the stronger the overall feature expression ability, and the corresponding extracted features are more abstract.

2) CAvg-Pooling and CMax-Pooling Comparison: This section conducts ablation analysis about CAvg-Pooling and CMax-Pooling on the SOC-10 and EOC-NV-10 datasets. The experimental results are shown in Table IX.

It can be seen from Table IX that on the SOC-10 dataset, the target recognition accuracy and loss using CMax-Pooling are 99.46% and 0.5563, respectively. The target recognition results after utilizing the CAvg-Pooling operation are Acc-99.79% and Loss-0.2848. Similarly, on the EOC-NV-10 (7.5 dB) dataset, MS-CVNets applying CAvg-Pooling can achieve better SAR target recognition performance, namely, Acc-98.56% and Loss-1.5658. Based on the above experimental analysis, it can be inferred that the implementation of CAvg-Pooling operation on the deeper target semantic features can effectively improve the target recognition performance of MS-CVNets.

3) Complex-Valued Feature Fusion Analysis: In MS-CVNets, how to effectively fuse the multiscale target semantic information and where to fuse it requires relevant experiments to verify and optimize. For this purpose, in this experiment, we design three different

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TABLE VIII

ABLATION ANALYSIS OF THE PROPOSED MS STRUCTURE WITH A COMBINATION OF DIFFERENT NETWORK CONFIGURATIONS

Datasets	Models	Kernel-3	Kernel-7	Kernel-11	Loss	Acc (%)
SOC-10	CVNet-3	✓	×	×	1.4541	98.68
	CVNet-7	×	✓	×	0.9229	99.22
	CVNet-11	×	×	✓	0.5767	99.38
	CVNet-3-7-11	✓	✓	✓	0.5196	99.46
	MS-RVNet	✓	✓	✓	0.6144	99.51
	MS-CVNets	✓	✓	✓	**0.2848**	**99.79**
EOC-NV-10 (SNR=7.5 dB)	CVNet-3	✓	×	×	15.6038	81.98
	CVNet-7	×	✓	×	10.4834	87.46
	CVNet-11	×	×	✓	12.6064	84.58
	CVNet-3-7-11	✓	✓	✓	38.8948	58.64
	MS-RVNet	✓	✓	✓	11.1928	86.68
	MS-CVNets	✓	✓	✓	**1.5658**	**98.56**

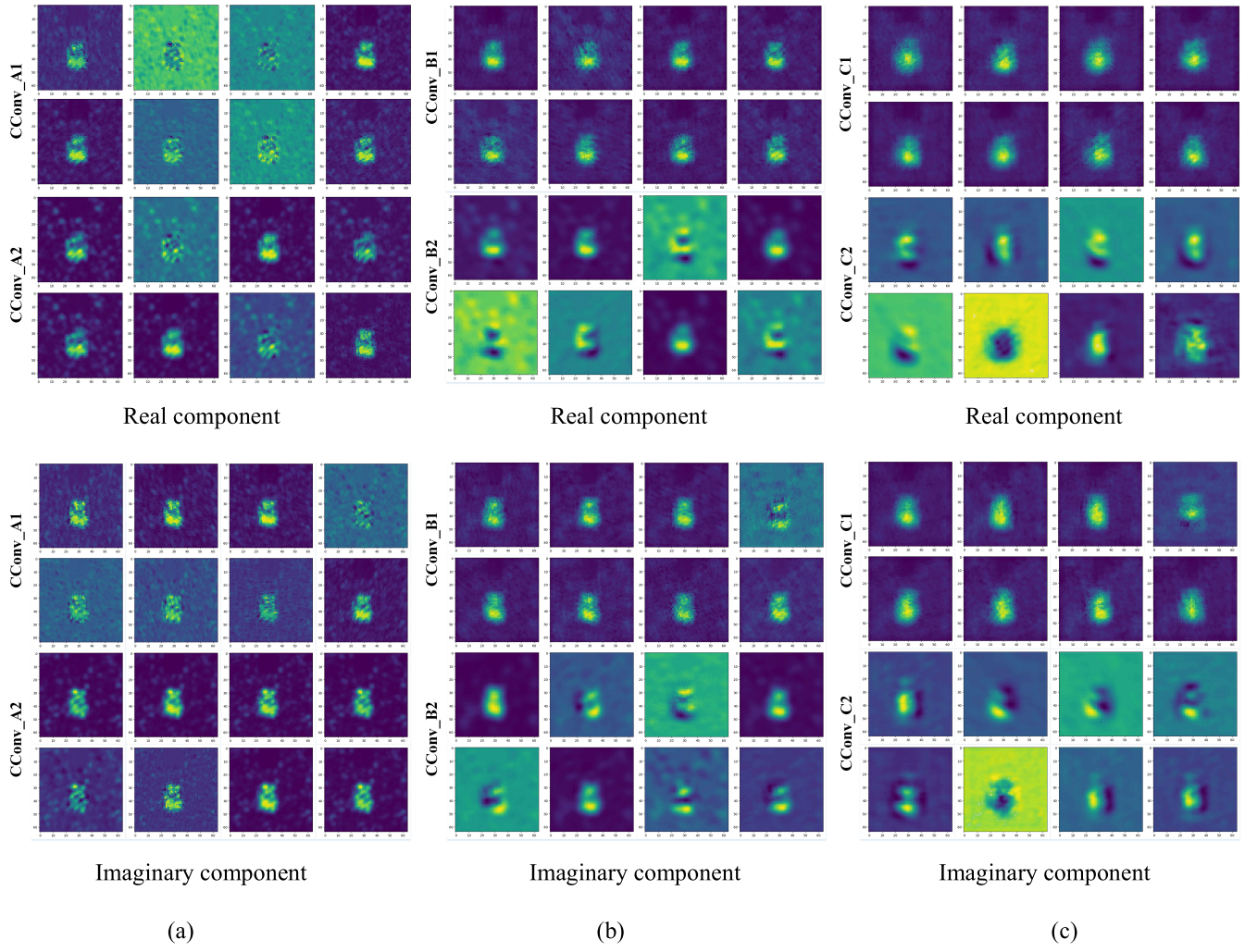

Fig. 13. MS features of MS-CVNets under different convolution kernel sizes. (a) Real and imaginary feature maps of the target T72 after the first layer of complex convolution block of Stream-A (kernel size-3). (b) Real and imaginary parts' feature map of the target T72 after the first layer of complex convolution block of Stream-B (kernel size-7). (c) Real and imaginary parts' feature map of the target T72 after the first layer of complex convolution block of Stream-C (kernel size-11).

feature fusion network architectures based on MS-CVNets: early fusion, middle fusion, and later fusion (Fig. 14), corresponding to MS-CVNets-1F-32, MS-CVNets-1F-16, and MS-CVNets-1F-4. For example, MS-CVNets-1F-32 indicates that the network model has only one feature fusion layer, and the feature size of the fusion layer is 32 × 32. Next, three feature fusion strategies, CAF, CMF, and CCF, are employed in MS-CVNets to verify their representation ability. The experimental results are delineated in Table X.

It can be seen from Table X that under the CCF feature fusion processing, the test loss and recognition accuracy results of different models are MS-CVNets-1F-32, 0.5153%–99.55%,

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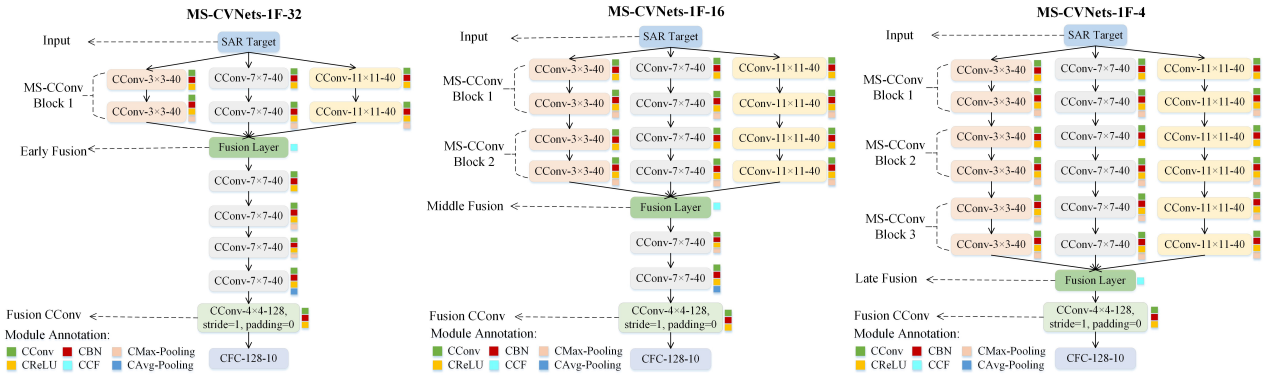

Fig. 14. Network architecture of early, middle, and late feature fusion. (a) MS-CVNets-1F-32 (early fusion). (b) MS-CVNets-1F-16 (middle fusion). (c) MS-CVNets-1F-4 (late fusion).

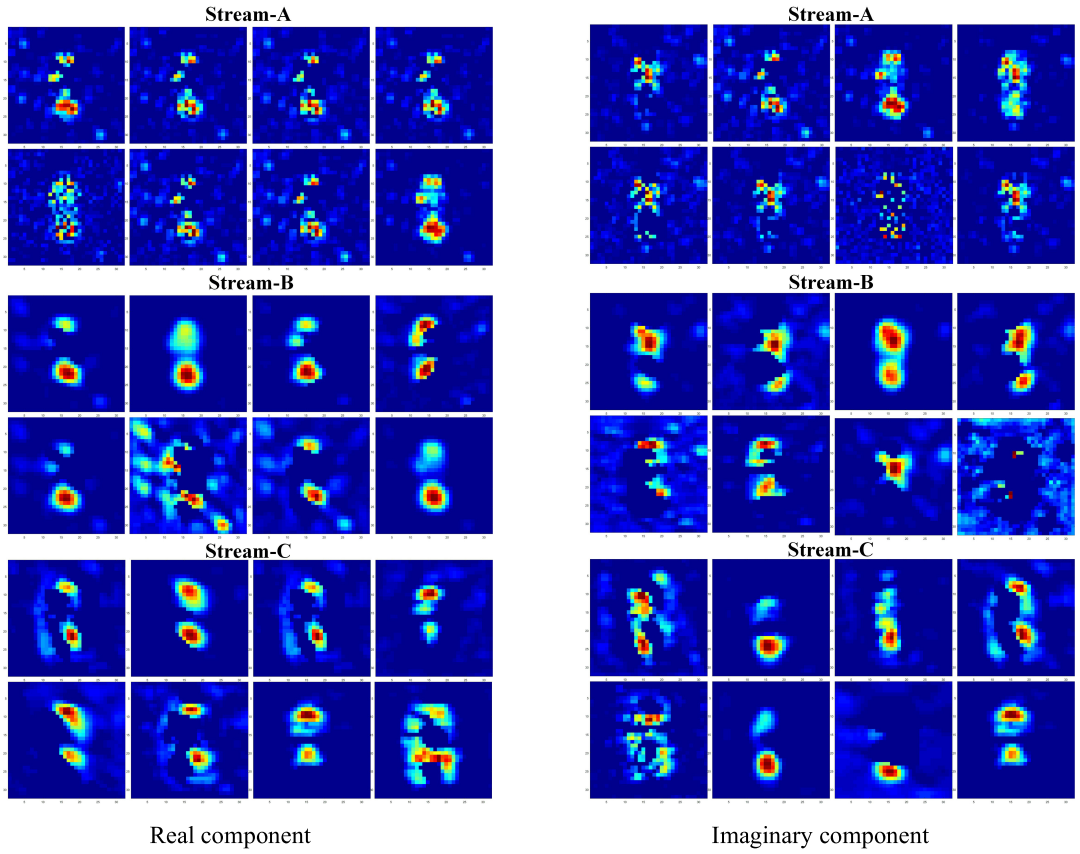

Fig. 15. MS complex-valued feature maps of MS-CVNets: Stream-A, Stream-B, and Stream-C.

TABLE IX
PERFORMANCE COMPARISON BETWEEN CAVG-POOLING AND CMAX-POOLING

Datasets	Models	CAvg-Pooling	CMax-Pooling	Loss	Acc (%)
SOC-10	MS-CVNets	×	✓	0.5563	99.46
	MS-CVNets	✓	×	**0.2848**	**99.79**
EOC-NV-10 (SNR=7.5 dB)	MS-CVNets	×	✓	32.2664	63.18
	MS-CVNets	✓	×	**1.5658**	**98.56**

MS-CVNets-1F-16, 0.5235%–99.51%, MS-CVNets-1F-4, 0.5926%–99.42%, MS-CVNets, and 0.2848%–99.79%. This indicates that the multiple feature fusion strategies (early, middle and late fusions) in MS-CVNets own

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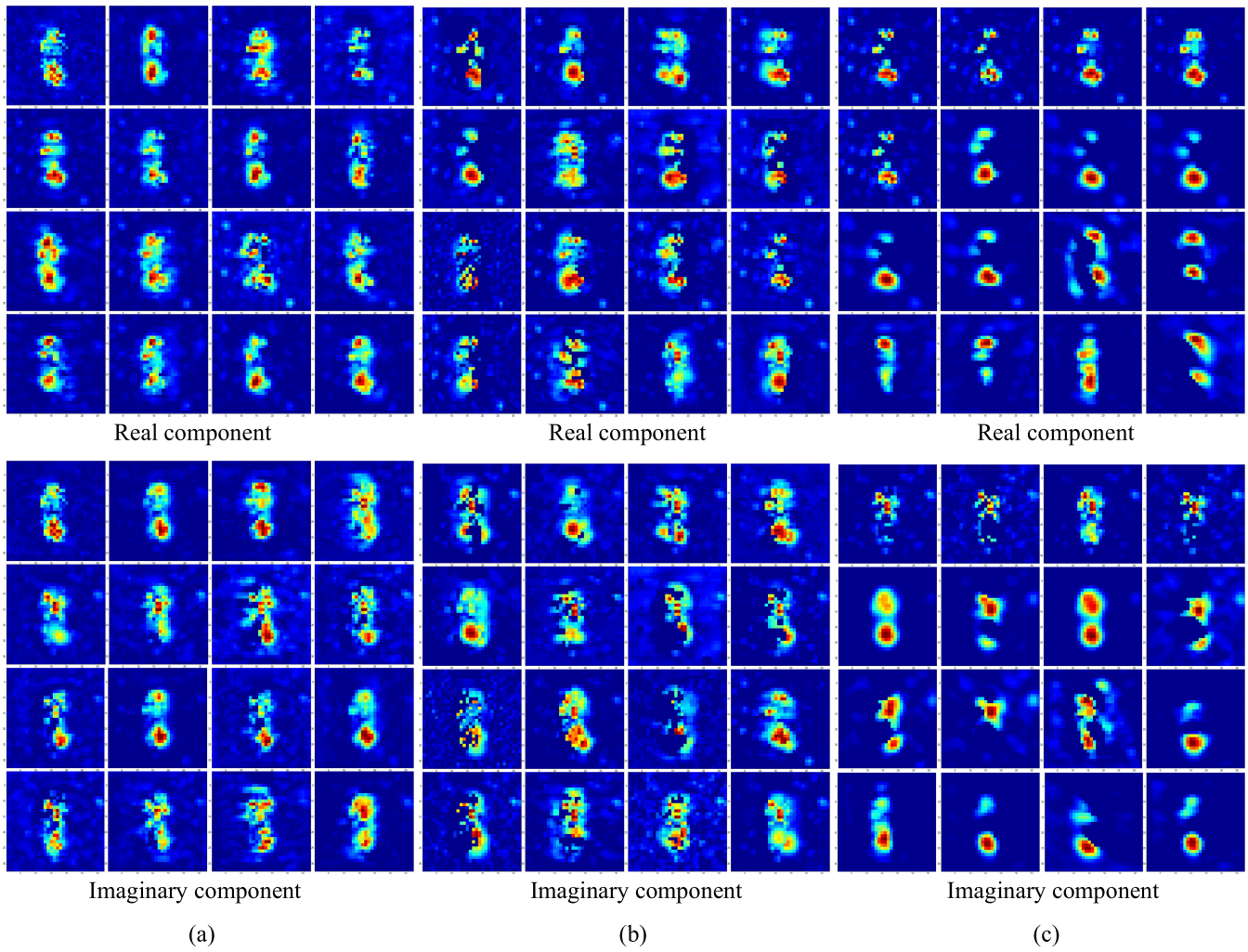

Fig. 16. Comparison on MS-CVNest MS features fusion. (a) CAF. (b) CMF. (c) CCF.

TABLE X
PERFORMANCE COMPARISON BETWEEN CAF, CMF, AND CCF

Datasets	Models	CCF	CAF	CMF	Loss	Acc (%)
SOC-10	MS-CVNets-1F-32	✓	×	×	0.5153	99.55
	MS-CVNets-1F-16	✓	×	×	0.5235	99.51
	MS-CVNets-1F-4	✓	×	×	0.5926	99.42
	MS-CVNets	×	✓	×	0.4511	99.59
	MS-CVNets	×	×	✓	0.5446	99.46
	MS-CVNets	✓	×	×	**0.2848**	**99.79**
	MS-CVNets-L2	✓	×	×	**1.0798**	**99.92**

better target recognition performance than single feature fusion layer. Moreover, comparing with CAF and CMF strategies, MS-CVNets applying CCF operation achieves the highest target recognition accuracy (99.79%), while the MS-CVNets model with L2 weight decay, MS-CVNets-L2, can further improve target recognition accuracy to the state of the art 99.92%, which demonstrates the powerful data fitting capabilities of MS-CVNets. For illustrative purposes, Figs. 15 and 16 reveal the feature maps of the extracted MS complex-valued feature after different fusion strategies.

As can be observed from Fig. 15 that different from the shallow convolution results, the target features extracted by the high-level convolution tend to be more macroscopic and integrated as the number of network layer increases. Meanwhile, according to the Streams-A–C feature maps, it is suggested that the CConv operations pay more attention to the local texture features of the target, such as the local intensity, edges, and shadows. The target features after CAF and CMF processing, as shown in Fig. 16(a) and (b), are richer and more complete, reflecting that CAF and CMF can effectively increase the target characteristics information of a single channel. Although

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the CCF operation does not have this characteristic, as shown in Fig. 16(c), it can acquire abundant multiscale target information by expanding the number of feature channels. To sum up, CAF, CMF, and CCF can effectively utilize the affluent multiscale target feature information in MS-CVNets and further provide an important guarantee for subsequent target recognition.

V. Conclusion

In this article, we propose an SAR target recognition method with MS feature fusion based on complex-valued calculations, called MS-CVNets, by integrating the complex-valued operations and the MS structure to effectively consider the phase information of SAR targets. A series of complex-valued building blocks, such as CConv, CBN, and CReLU, are designed for the purpose of constructing CVNets at arbitrary scale levels. Benefiting from the MS structure, the proposed MS-CVNet is capable of extracting rich multiscale features of SAR targets. Moreover, through simulating the complex-valued calculation mechanism, three well-designed complex-valued feature fusion strategies, CAF, CMF, and CCF, are released to improve the representation ability of the proposed MS-CVNets. Experimental results on the MSTAR dataset illustrate that the proposed MS-CVNets can achieve better target recognition performance than the state-of-the-art real-valued models. In the future, we will focus on more lightweight and faster complex-valued SAR target recognition methods to enhance the computational efficiency.

Appendix

This section is mainly discussed the use of bias in the definition of CFC layers. Normally, the biases are defined for each set of weights. Therefore, it would expect to see that the bias b_r is used for O_{rr} and O_{ri} and b_i for O_{ii} and O_{ir}. However, we only define the bias, b_r and b_i, to O_{rr} and O_{ir}, respectively, for the purpose of reducing parameters.

Suppose we define corresponding b_r and b_i to all O. We have

$$\begin{cases} O_{rr} = \mathbf{w}_r \cdot \mathbf{x}_r + b_{rr} \\ O_{ii} = \mathbf{w}_i \cdot \mathbf{x}_i + b_{ii} \\ O_{ri} = \mathbf{w}_r \cdot \mathbf{x}_i + b_{ri} \\ O_{ir} = \mathbf{w}_i \cdot \mathbf{x}_r + b_{ir}. \end{cases} \tag{34}$$

According to the calculation formulation of CFC

$$\begin{aligned} O_{\text{CFC}} &= (O_{rr} - O_{ii}) + i \cdot (O_{ri} + O_{ir}) \\ &= (w_r \cdot x_r - w_i \cdot x_i + (b_{rr} - b_{ii})) \\ &\quad + i \cdot (w_r \cdot x_i + w_i \cdot x_r + (b_{ri} + b_{ir})) \end{aligned} \tag{35}$$

where $(b_{rr} - b_{ii})$ and $(b_{ri} + b_{ir})$ are constants, corresponding to the bias, b_r and b_i, of the real and imaginary parts, respectively. Combining it together, then

$$O_{\text{CFC}} = (w_r \cdot x_r - w_i \cdot x_i + b_r) + i \cdot (w_r \cdot x_i + w_i \cdot x_r + b_i). \tag{36}$$

Therefore, in order to save the amount of parameters, we use b_r in O_{rr} and b_i in O_{ir} for the equivalent calculation

$$\begin{cases} O_{rr} = \mathbf{w}_r \cdot \mathbf{x}_r + b_r \\ O_{ii} = \mathbf{w}_i \cdot \mathbf{x}_i \\ O_{ri} = \mathbf{w}_r \cdot \mathbf{x}_i \\ O_{ir} = \mathbf{w}_i \cdot \mathbf{x}_r + b_i. \end{cases} \tag{37}$$

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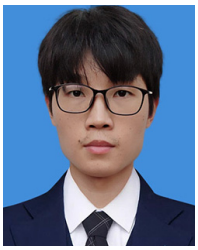

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